

Risk Measure, Risk Aversion, Utility Function

Investors generally want to avoid risk.

(magnitude of risk).

In finance, standard deviation is the most common measure of risk.

It quantifies the volatility or fluctuation of an investment's return;
a higher standard deviation indicates greater variability and, consequently, higher risk.

An utility function is used to reflect an individual's specific risk aversion tendency.

Portfolio	Expected Return	standard deviation
A	5%	9%
B	8%	15%
C	12%	20%

Mean - Variance Utility Function

$$U = E(r) - \frac{1}{2} A \sigma^2$$

U : Mean-Variance utility value.

$E(r)$: The expected return of the portfolio

A : Risk aversion coefficient

σ^2 : Variance of the returns

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- If $A > 0$: The investor is Risk Averse
 - If $A = 0$: The investor is Risk Neutral
 - If $A < 0$: The investor is Risk Seeking

Mean - Variance Criterion

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"Portfolio A dominates Portfolio B" if the following conditions are met:

- $E(r_A) \geq E(r_B)$ (A has equal or higher return)
- $\sigma_A \leq \sigma_B$ (A has equal or lower risk)
- At least one of these inequalities must be strict.

